

Kendy Project: Docker Deployment Architecture Report

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Abstract

This document provides a comprehensive overview of the Docker-based deployment architecture for the Kendy application. It details the containerized services, their interactions, network configuration, and operational procedures. The report serves as both technical documentation and operational reference for development and production environments.

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1 System Architecture

1.1 Container Architecture

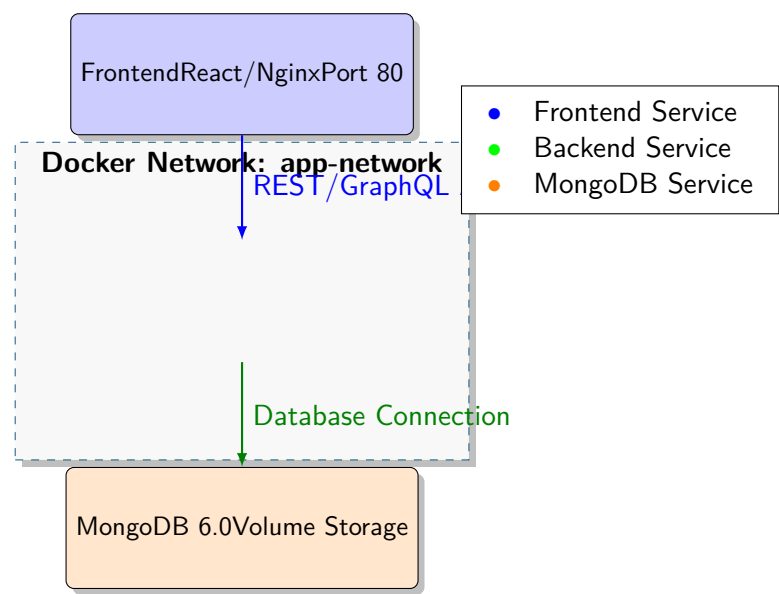


Figure 1: Container Architecture Diagram showing service isolation and communication paths

1.2 Service Specifications

Component	Configuration Details	Port
Frontend	- React app with Node.js 20 - Served via Nginx Alpine /health endpoint - Static assets with gzip	80
Backend	- Node.js 20 runtime - Volume: /uploads - JWT authentication - MongoDB connection pooling /api/health endpoint	5001
MongoDB	- MongoDB 6.0 image - Volume: /data/db - Init via entrypoint - WiredTiger engine	27017

Table 1: Detailed service configurations and technical specifications

2 Operational Characteristics

2.1 Service Dependencies and Startup Sequence

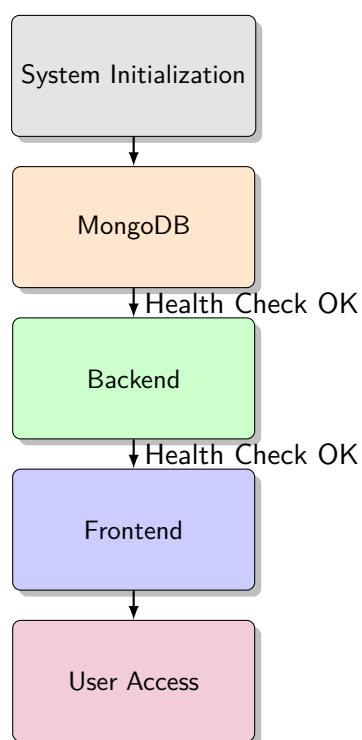


Figure 2: Service dependency graph showing startup sequence with health verification

2.2 Network Traffic Flow

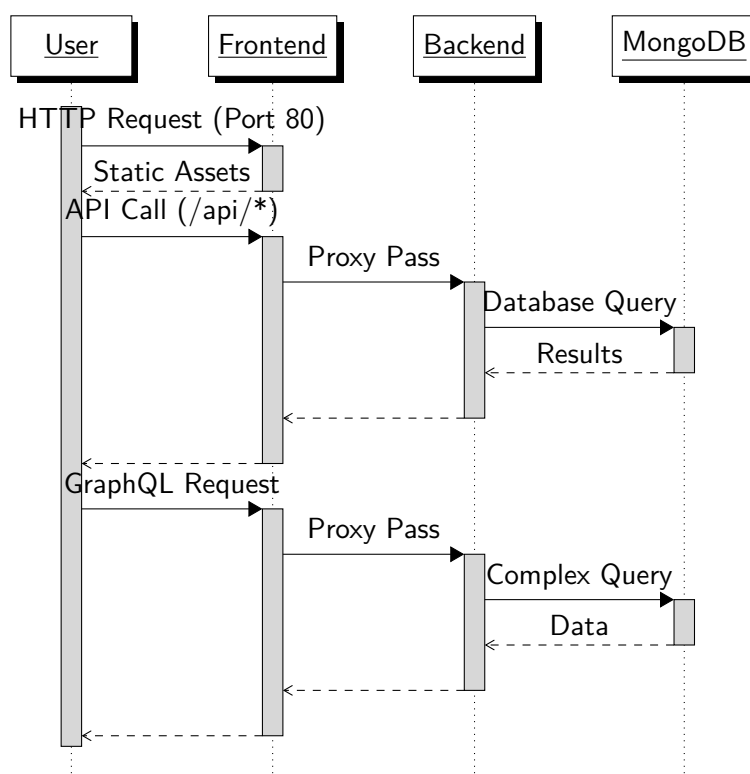


Figure 3: API request processing sequence with database interaction

3 Infrastructure Configuration

3.1 Network Configuration

- **Bridge Network:** Custom app-network
- **Port Exposure:** Only frontend (80) exposed
- **Service Discovery:** Container names
- **Gateway:** Internet access

3.2 Volume Management

- MongoDB: persistent volume
- Backend: upload volume
- Host backups

4 Key Design Decisions

- **Isolated Services:** one container per role
- **Health Monitoring:** readiness endpoints
- **Efficiency:** Alpine images
- **Security:**
 - Non-root user
 - Network isolation
 - Minimal exposure

- itemize

5 Operational Procedures

5.1 Deployment Commands

```
1 docker compose up -d --build
2 docker compose ps
3 docker compose logs -f
4 docker compose down -v
```

5.2 Database Administration

```
1 docker exec -it kendy-mongo-1 mongosh
```

```
1 db.users.find().pretty()
2 db.users.drop()
3 use KendyProject
4 db.dropDatabase()
```

5.3 Reset Environment

```
1 docker compose down
2 docker compose up -d
```

5.4 Docker Cache Maintenance

```
1 sudo docker system prune -a
```

5.5 Rebuild Without Cache

```
1 docker compose build --no-cache
```

5.6 Backup Database

```
1 docker exec -i kendy-mongo-1 mongodump --db KendyProject --out /data/backup
2 docker cp kendy-mongo-1:/data/backup ./backup
```

5.7 Restore Database

```
1 docker cp ./backup kendy-mongo-1:/data/backup
2 docker exec -i kendy-mongo-1 mongorestore --db KendyProject /data/backup/KendyProject
```

5.8 Copy to USB (Optional)

```
1 cp -r ~/Kendy/backup/KendyProject /media/med/MaCleUSB/
```

6 Conclusion

The Dockerized architecture provides a robust, scalable foundation for the Kendy application.